Fiscal Year:	FY 2019	**Task Last Updated:**	FY 11/15/2018
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	GeneLab: Revealing Spaceflight- and Gravity-Response Networks in Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-GeneLab. Appendix A: Translational Systems Biology and Informatics Research Using the GeneLab Data System
Start Date:	12/19/2016	**End Date:**	12/18/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	25	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/18/2019 per NSSC information (Ed., 9/24/19)		
Key Personnel Changes/Previous PI:	November 2018 report: None		
COI Name (Institution):	Barker, Richard Ph.D. (University of Wisconsin)		
Grant/Contract No.:	NNX17AD52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The GeneLab data repository is a powerful resource with which to dissect plant responses to spaceflight at the molecular and genetic level. The research in this GeneLab Innovation study therefore seeks to continue and extend our current analysis of the GeneLab plant science data through bioinformatics, experimental tests of predictions, and development of approaches to facilitate data exploration and rapid sharing of analytical tools. This work directly addresses Goal 1 of the Space Biology program by creating 'new knowledge of how biological systems adapt to space.' In addition, this research approaches some of GeneLab's guiding questions as to how implementation of large scale dataset analyses can help produce new insights and hypotheses about the effects of spaceflight on biological systems. Aim (1): We are using informatics to ask how spaceflight alters the plant transcriptome and proteome using GeneLab data. We are then comparing these changes to responses to other stresses and environments, (e.g., pathogen attack or light stress) where large-scale, publicly available datasets have already been deposited outside of GeneLab. These datasets include responses to partial gravity and gravitropism-related transcriptomes. Integrating such analyses with allied data, such as QTL maps, will allow us to specifically ask how many of the spaceflight-linked changes may be mediated through altered gravity sensing and response. Our preliminary analysis of Arabidopsis data from BRICs (Biological Research in Canisters)16-20 has revealed oxidative stress and hypoxia as major flight-related stresses and has targeted gene families ranging from heat-shock proteins to cell wall peroxidases as likely key players in this spaceflight response network. The more comprehensive analysis being undertaken in this innovation study will greatly strengthen these initial findings and expand the range of molecular mechanisms we can detect. Aim (2): The second element of the work is to take the genes defined from the informatics in Aim 1 and test their function by passing their mutants through a suite of phenotypic assays. Analyses includes morphometric measurements of gravtiropic, thigmotropic, and clinostat growth in addition to development in short and long-term hypoxia. These studies are being performed using automated image analysis tools such as RootNav, RootTrace, and custom R-Shiny apps that are highly compatible with rapid sharing of data and analyses via the cloud. Inferences from these growth analyses will be compared to those revealed by quantitative PCR results of marker genes identified from the core spaceflight, oxidative stress, hypoxia, and gravitropism transcriptomic analyses. Aim (3): GeneLab datasets also offer the opportunity to address the key challenge of facilitating exploration of these large, complex, and specialized datasets by non-specialists. The Gilroy lab has a long-standing collaboration with iPlant/CyVerse and Cold Spring Harbor in the development of RNAseq analysis teaching tools such as the DNAsubway Green Line and the Cold Spring Harbor DNA learning center. Extending these approaches to the spectrum of GeneLab's plant data will make a significant step towards promoting access and understanding. One further approach to increasing accessibility for the non-specialist that we are developing will be through an online Tableau or Qlik-based app. These tools build highly interactive data visualization that allows non-expert users to search across, and explore interconnections within, large and complex datasets. The initial data visualization tool we have developed contains all GeneLab Arabidopsis data but is designed to be capable of integrating with other datasets to greatly extend the reach of the GeneLab resources.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is leading to benefits outside of the spaceflight arena at two levels: (1) tool development and (2) extension of basic knowledge of plant stress responses. At the level of tool development, we have generated a large dataset exploration environment built around the Qlik data visualization toolset. This tool allows rapid comparison and exploration between large-scale 'omics-level datasets, facilitating comparison between transcriptomics/gene expression data and then cross-comparisons to protein-level data such as proteomics analyses. Although focused on spaceflight data within GeneLab's data repository, this architecture is applicable to wider experimental datasets. In addition, we are developing 3-d printing technologies to rapidly build plant analysis stations with a focus on large parallel arrays of clinostats and imaging stations allowing phenotyping of gravity-related effects on plants. In parallel we have also been developing gas regulation chambers to allow analysis of plant growth and molecular phenotypes related to lowered oxygen availability. Although again focused on spaceflight-related events of lowered oxygen levels, these devices are also applicable to studying the effects of ground-based phenomena such as flooding effects on plant biology. Our analyses of phenotypes has also extended basic understanding on plant responses to touch, low oxygen levels, and oxidative stress signaling on Earth. For example, we have been able to define an array of genes that when mutated alter responses to oxidative stress on Earth. Oxidative stress is a major element of stress signaling and resistance pathways and so this basic understanding is expanding our broad knowledge of how plants are able to respond to less than optimal environmental conditions.
	We have developed, and now deployed, a data exploration environment for mining the GeneLab plant biology datasets and combining them with other available spaceflight 'omics-level' analyses using the Qlik data visualization system. We have imported all the GeneLab plant biology data alongside publicly available datasets from a range of researchers into a database built around the Qlik platform for data visualization. The system is named the Test Of All Space Transcriptomes, or TOAST, database. In parallel, we have imported publicly available datasets related to spaceflight-associated stresses including: hypoxia, oxidative stress, and radiation exposure to facilitate cross-referencing to the GeneLab datasets. The RNAseq datasets have been imported as raw data and reanalyzed using the Tuxedo pipeline to ensure uniformity in data structure and comparability. Opening a dataset allows interactive browsing of the transcriptome data and real-time filtering of genes on parameters such as significance of fold-changes induced in spaceflight and Gene Ontology annotations. The database has been in public beta for the last year allowing us to improve the user interface, increase the coverage of plant biology datasets, and increase the stability and overall end user experience in line with suggestions from the beta-testing research and educational community. A second major goal for this work is to use phenotyping of mutants in the plant genes identified as showing strong links to spaceflight-related events from the TOAST-based analysis described above. This phenotyping is being used to ask how well TOAST generates reliable and testable hypotheses about plant responses to spaceflight. Gravitropism, reaction to hypoxia, and response to oxidative stress are closely linked to spaceflight-related conditions due to spaceflight's disruption in normal plant gravity signaling, reduced availability of oxygen from the lack of convective mixing of gasses in space, and high oxidative loads due to a general background of spaceflight-related stress. Analyses from the TOAST

Task Progress:	database have identified Heat Shock Proteins (molecular chaperones) and related genes such as the Bcl-2 associated athanogene family (BAG family of chaperone regulators) and cell wall peroxidases as likely key players in spaceflight stress responses. Phenotyping mutants in these genes is revealing a range of spaceflight-related phenotypes including some unexpected roles for the heat shock proteins in gravitropic response and the BAG genes in low oxygen stress responses. In addition to a research discovery tool, we have also tested the TOAST database in the educational setting. For example, we have used TOAST at the Middle School level, where students went through gene discovery and hypothesis generation. Two of these students then stayed on to develop a major report on the bioinformatics they performed and are working with the lab to translate the in silico work to wet-lab based experimentation. In addition, we have incorporated TOAST into an undergraduate class experience at Carthage College, a 4 year liberal arts college in Wisconsin. In collaboration with Dr. Andrea Henle, TOAST analyses were incorporated into her Astrobiology undergraduate class with the goal of leading the students through an authentic research experience, discovering new genes related to spaceflight response in Arabidopsis. We also maintain a “Collaboratory” where biology and engineering undergraduates and high school students come together to develop high throughput phenotyping equipment and other hardware relevant to our spaceflight-related work. Over 2017-2018 we have mentored approximately 10 independent study students working on various plant molecular and engineering projects related to space biology and GeneLab. We also run a space biology related practical lab course called AstroBotanical Engineering. Here a pool of plant biology and engineering students collaborate to develop space-related hardware for ground-based testing of plant science space-related projects. We have ~15 students per semester. These students are helping develop the phenotyping systems to test the predictions made through using the GeneLab analyses from the TOAST database. This year we have concentrated on space radiation-related changes and defining target genes for future radiation experiments. During 2017-2018, we have presented GeneLab-related research at the 2018 MidWest Plant Cell Dynamics Meeting, the annual meeting of the American Society for Gravitational and Space Research, and the American Society of Plant Biologists, and at international venues such as the Academia Sinica in Taiwan. GeneLab analyses are also key elements of our local outreach efforts, where we have presented this work at events ranging from University of Wisconsin sponsored outreach days (e.g., UW’s Science Saturdays and Science Expeditions) to presentations for high school students and undergraduates (e.g., BioHouse) and K-12 teachers (Biotechnology Institute summer training program), and are now hosting a BTI intern for the coming year to work on space transcriptome work. GeneLab analyses are also integral to a middle/high school ‘field trip’ as part of a collaboration with the Wisconsin Institutes for Discovery called “plants and robots.” Here ~15-20 students are introduced to plant space science and then program a robotics imaging platform to perform plant gravitropism assays. We support 10-20 of these field trips over the year. We also maintain a popular science outreach site called Astrobotany.com (https://) that promotes the latest in space-related plant biology. Astrobotany.com also hosts a tutorial for TOAST and acts as one portal to the TOAST database. We are also working closely with a game development studio at the Wisconsin Center for Education Research within the University of Wisconsin to develop an educational video game built around colonizing Mars. We are using the GeneLab datasets as the background data driving the plant biology aspects of the game. This is currently in the initial stages of development and we are engaging the Boys and Girls Club of Dane County to bring in a cohort of students to help develop concepts and initial programming for the game. In addition, we have presented this work at a range of outreach event such as Science on Tap and at Madison’s Space Place, where the talk was recorded for release through Wisconsin Public Television. We have had the opportunity to extend the impact of the work with some national radio exposure, such as WPR’s Central Time and NPR’s Science Friday and on The Big Ten TV network. We have combined these national efforts with international outreach, such as lab members working with Seva Chakkara, an orphanage of ~100 children in India.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Hilleary R, Gilroy S. "Systemic signaling in response to wounding and pathogens." Curr Opin Plant Biol. 2018 Jun;43:57-62. Epub 2018 Jan 17. Review. https://doi.org/10.1016/j.pbi.2017.12.009 ; PubMed PMID: 29351871 , Jun-2018
Articles in Peer-reviewed Journals	Hilleary R, Choi WG, Kim SH, Lim SD, Gilroy S. "Sense and sensibility: the use of fluorescent protein-based genetically encoded biosensors in plants." Curr Opin Plant Biol. 2018 Dec;46:32-8. Review. Epub 2018 Jul 21. https://doi.org/10.1016/j.pbi.2018.07.004 ; PubMed PMID: 30041101 , Dec-2018